



An integrated approach for warehouse analysis and optimization: A case study



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ARTICLE INFO

Article history:

Received 14 June 2013

Received in revised form 11 June 2014

Accepted 10 December 2014

Available online

Keywords:

Warehouse

Analysis

Optimization

Value Stream Mapping

Unified Modeling Language

Genba Shikumi

ABSTRACT

The paper focuses on the analysis and optimization of production warehouses, proposing a novel approach to reduce inefficiencies which employs three lean manufacturing tools in an integrated and iterative framework. The proposed approach integrates the Unified Modeling Language (UML) – providing a detailed description of the warehouse logistics – the Value Stream Mapping (VSM) tool – identifying non-value adding activities – and a mathematical formulation of the so-called Genba Shikumi philosophy – ranking such system anomalies and assessing how they affect the warehouse. The subsequent reapplication of the VSM produces a complete picture of the reengineered warehouse, and using the UML tool allows describing in detail the updated system. By applying the presented methodology to the warehouse of an Italian interior design producer, we show that it represents a useful tool to systematically and dynamically improve the warehouse management. Indeed, the application of the approach to the company leads to an innovative proposal for the warehouse analysis and optimization: a warehouse management system that leads to increased profitability and quality as well as to reduced errors.

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1. Introduction

Nowadays, factors like globalization, competition, rapid market changes, short product life cycles, high productivity and reduction of time-to-market make the impact of logistics on production by far wider than in the past [1–5]. Such a complex scenario has led to a considerable interest in the design, planning and control of warehousing systems as new research topics ([6–10], Gu et al., 2010; [11,12]). Warehouses may be defined as material handling stations dedicated to receiving, storage, order-picking, accumulation, sorting and shipping of goods [13]. This definition covers a wide variety of systems, and may be broadly categorized by three types of warehouses [14]: (i) distribution warehouses, (ii) production warehouses and (iii) contract warehouses.

This paper considers production warehouses, a class that covers the vast majority of warehouses and may be defined as production

sites facilities for storing raw materials, semi-finished products, and final products [14]. Moreover, considering the traditional classification of strategic, tactical, and operational problems, we focus on the strategic design of production warehouses, that has been shown to be a sector in need for investigation, as opposed to the strong research oriented on single sub-problems at an operational level that seems to be dominant in the literature [11].

The available literature on production warehouse strategic design and performance evaluation/optimization may be classified based on three main issues [15,16]: (i) contributions addressing warehouse design decisions, typically concentrating on order picking routing or product location strategies; (ii) papers proposing analytic or simulation models and (iii) publications on benchmarking and performance evaluation. The interested reader may refer to the comprehensive reviews by Baker and Canessa [17], Gu et al. [16] and Rouwenhorst et al. [11] for a discussion on the first two topics. In spite of the importance of warehouse design and their management, the technical literature on warehouse performance assessment is meager, with two main streams [15]: (a) contributions proposing frameworks for designing or analyzing warehouses and (b) those which directly address performance assessment. The reader interested in the latter topic is addressed to

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[15], where the authors tackle the warehouse coordination problem by analyzing the warehouse as a single system and quantifying its performance as a whole by a data envelopment analysis approach. On the contrary, many authors agree on the lack of systematic approaches on warehouse analysis and optimization [17,18]. This paper fills this gap proposing an integrated approach for warehouse analysis and optimization which employs several tools borrowed from lean manufacturing.

Lean manufacturing, originating from the Toyota production system, is the most widely known approach for industrial improvement [19]. Basically, the major goal of lean production is to reduce costs by eliminating any possible source of waste within the company. The seven sources of waste identified by the lean manufacturing are: over-production, waiting time, transportation, over-processing, inventory, motion, and scrap [20]. Hence, lean manufacturing is a multi-dimensional approach which comprises a variety of management practices and tools [21]. However, these tools are not integrated but are usually employed on an isolated basis. Moreover, they are typically applied to improve the internal logistics of the company and not the warehouse.

The research community has only recently started the application of lean principles to the warehouse [22,23]. Among the lean manufacturing tools, one of the most commonly cited in the related literature is Value Stream Mapping or VSM [24], thanks to its ability to provide a compact display of the system under examination. For this reason, there are several application examples of VSM to discrete manufacturing systems (see e.g. [25–27]). However, only few papers employ VSM for lean warehousing (see for instance [28–30]). Moreover, these contributions do not consider VSM in an integrated fashion with other tools for the systematic analysis and optimization of the warehouse.

Summing up only the integration of lean tools allows to analyze the overall warehouse management problem from different point of view solving it in a more logical approach.

Based on the lack of joined tools found in the related literature, this paper proposes a lean warehousing approach for analyzing and reducing waste in production warehouses. The presented methodology integrates the Unified Modeling Language (UML) formalism, the VSM lean manufacturing tool, and the so-called Genba Shikumi philosophy. In particular, the UML formalism is adopted to describe and analyze the warehouse activities and the corresponding responsibilities; this detailed representation is employed by the VSM tool to highlight in which area any possible waste is located; hence, the Genba Shikumi philosophy is adopted to rank the analyzed waste and determines the most significant ones to eliminate. This scope is obtained by reapplying the VSM and UML tools to respectively redesign the system without the described anomalies and detail its flows and activities. The physical implementation of the designed changes follows in the so-called final reengineering phase.

The iterative nature of the presented integrated approach brings to the so-called “kaizen”, that is the continuous improvement which the lean philosophy is based on.

To enlighten the methodology, we employ a real case study, an Italian producer of interior design objects. The proposed technique is applied to improve some key performance indicators (KPI) of the company production warehouse, enlightening the approach effectiveness and leading to an innovative proposal for the warehouse optimization, with particular reference to the key order picking process.

The remainder of the paper is structured as follows. Section 2 presents an overview of lean manufacturing techniques motivating the presented approach, which is described in detail in Section 3, where its merits are also discussed with respect to the related literature. Hence, Section 4 presents the case study in order to

validate the presented approach and discusses the obtained results, describing a possible reengineering proposal and the consequent future mapping of the case study. Subsequently, Section 5 presents the conclusions and possible future developments. The paper is concluded by the reference section.

2. Summary of lean manufacturing techniques and paper motivation

Lean manufacturing includes numerous tools and methodologies [31]. Among these, we recall those somehow connected with the methodology proposed in this paper.

1. VSM – VSM [21] is a tool used to visually map the flow of production and to show the current and future state of processes in a way that highlights opportunities for improvement. Dharmapriya and Kulatunga [28] apply VSM to optimize warehouse operations and use the simulated annealing heuristics to determine an optimal layout. Moreover, Gopakumar et al. [29] employ discrete event simulation and VSM for studying the dock allocation problem in a food distribution center, while Chen et al. [30] investigate the use of VSM with radiofrequency identification in the warehouse.
2. Genba – Genba [32] is a philosophy that recommends managers to spend their time on the plant floor (rather than in their office), i.e., in the place where real action occurs; it is a companion approach of VSM that focuses on the waste identification and removal, i.e., the ideal system, rather than on the actual one.
3. Jidoka – Jidoka [33] consists in partially automating the manufacturing process (note that partial automation is typically much less expensive than full automation) and automatically stopping when defects are detected.
4. Bottleneck analysis – it consists in identifying which part of the manufacturing process limits the overall throughput and improve the performance of that part of the process [34].
5. Kaizen – it stands for a proactive strategy for achieving continuous improvement [35].
6. Key performance indicator (KPI) evaluation – it is a technique related to kaizen that consists in the continuous analysis and monitoring of metrics designed to track and encourage progress toward critical goals of the organization [36].
7. Muda removal – it consists in the reduction or elimination of anything in the manufacturing process that does not add value from the customers’ perspective [37].
8. PDCA – PDCA (Plan, Do, Check, Act) is an iterative methodology for implementing continuous improvement of processes and products [38].
9. Poka-Yoke – it is the design error detection and prevention into production processes with the goal of achieving zero defects [39].
10. Root cause analysis – it is a problem solving methodology that focuses on resolving the underlying problem instead of applying quick fixes that only treat immediate symptoms of the problem [40].
11. Visual factory – it consists in the utilization of visual indicators, displays, and controls throughout manufacturing plants to improve the communication of information [41].
12. Responsibility based planning and control – it is the practice of involving engineers in the planning process in order to ensure deadlines are met, backup plans are available, and tasks are completed [42].

It is to be noted that all these lean tools are rarely integrated into a streamlined, high quality system [43]. Recently, the

application to lean manufacturing of information and communication technologies (ICT) is attracting the interest of researchers and the reader is addressed to Mo [44], Riezebos and Klingenberg [45], Riezebos et al. [46], and Powell et al. [47] for some discussions on the topic. In such a lean and ICT change scenario, companies are involved in their entirety with all their different areas, but a decisive role is played by internal logistics and particularly by warehouse management, which is aimed at optimizing the flow of material and information within the company to maximize profit [48]. In this sense, warehouses provide useful services, such as the possibility to quickly respond to demand changes or to reduce transportation costs and to provide better customer services. Nevertheless, warehouse analysis and optimization is typically carried out in a traditional context, e.g., by using simulation [18], eventually in an uncertain environment [49]. Moreover, the existing contributions do not typically consider the problem of warehouse design in a continuous improvement context. On the contrary, with the enhanced customer demand and global competition, many manufacturing industries have to continuously monitor and progress the warehouse process. Indeed, although the manufacturer's main objective is obviously to minimize their production costs, there are numerous non-value adding activities like warehousing that contribute to create additional costs in the overall production. Furthermore, the main customer requirements are to get the desired goods on time, at the lowest price and in good condition. Therefore, in order to meet the customer and manufacturer perspectives, warehouse operations have to be optimized, eliminating inefficiencies and making them reliable in terms of cost. Hence, the importance of warehouse design and continuous improvement arises in industrial management and control. In the sequel we present an approach addressing such an emerging need for implementing lean operations in warehouses.

3. Overview of the proposed approach

The proposed warehouse analysis and optimization approach is aimed at reducing inefficiencies in warehouses employing lean manufacturing tools in an integrated and iterative framework. More in detail, we reengineer the company's product warehouse proposing a novel lean warehousing strategy that integrates the UML formalism [50] to describe in detail the internal logistics and the warehouse activities, and the VSM lean manufacturing tool as a simple and effective approach to identify and reduce the system anomalies (i.e., waste or *muda* in the Japanese term) in the value stream with particular reference to the warehouse, while improving the overall operational control. Moreover, if some *muda* are detected, the Genba Shikumi approach (the Japanese for engineering method) [51] is used to rank their significance. Based on the Genba Shikumi results, the VSM approach is hence adopted to establish the future vision of the value stream, where the impact of non-value added activities is strongly reduced. Hence, the UML tool is employed to suitably detail the updated warehouse, removing the main system anomalies previously detected, and implementing the designed changes. The implementation of the reengineering of the warehouse takes then place.

The proposed methodology is visually represented in Fig. 1, which describes all the previously summarized steps composing the approach. The approach is iterative, in order to ensure that improvement may be continuously achieved, in accordance with

the lean philosophy perspective. In the following we detail each step of the approach.

3.1. UML: describe and detail the warehouse

According to the first step of the approach in Fig. 1, we use UML to represent in detail the system activities. Indeed, UML is a standardized general-purpose modeling language, accepted in the year 2000 by the International Organization for Standardization (ISO) as an industrial standard. It includes a set of graphical tools to specify, visualize, modify, construct and document a system; it combines techniques derived from data modeling (entity relationship diagrams), business modeling (work flows), object modeling, and component modeling. UML can be used with almost any processes and across different implementation technologies, offering a standard way to visualize a system's architecture blueprints, including elements such as activities, actors, business processes, database schemas, logical components [50]. UML diagrams represent two different possible views of a system model [52]: (1) a structural view, emphasizing the static structure of the system using objects, attributes, operations and relationships by means of class diagrams and composite structure diagrams; (2) a behavioral view, emphasizing the dynamical behavior of the system and showing collaborations among objects and changes to the internal states of objects by means of sequence diagrams, activity diagrams and state machine diagrams. In particular, UML activity diagrams are similar to flowcharts but also allow representing a parallel processing in order to explain the workflow of the system and the critical points in each activity, pointing out all the possible paths, parallel processes, and their subdivisions and showing the actor and responsibility of each activity, hence providing also a dynamic view of the system.

The main elements to build these diagrams are the following: (1) the initial and final activities, respectively represented by a solid black circle and a bull's-eye symbol (an encircled black circle); (2) activities, represented by rounded rectangles; (3) arrows, that run from the start toward the end and represent the order in which activities happen; (4) forks and joins, represented by a thick bar, where forks refer to actions whose execution begin simultaneously, while joins indicate that all incoming actions must end before the flow proceeding; (5) decisions, i.e., alternate paths, represented by diamonds with options on each arrow; (6) swim lanes dividing the diagram in columns, expressing responsibilities; (7) numbers in and out of the flow, that refer to another diagram in which the individual activities between them are further detailed.

3.2. VSM: identify the anomalies

In this section, according to the second step of the lean warehousing approach shown in Fig. 1, all the problems and anomalies occurring in the warehouse are identified, with the aim of determining the sources of errors and inefficiencies and removing them. The main purpose is to describe all the existing interactions between the parts composing the warehouse processes, in order to standardize and optimize them. To this aim, the proposed lean warehousing approach uses the VSM technique, a fundamental tool of the so-called lean philosophy.

VSM is a lean tool used to analyze and design the flow of materials and information required to bring a product or a service



Fig. 1. A scheme of the proposed approach for warehouse analysis and optimization.

Table 1

Example of muda matrix and vector.

#	Observed problems	Waste 1	Waste 2	[...]	Waste w	MV
1	Problem 1					1
2	Problem 2	1			1	2
...	[...]		1			1
p	Problem p		1	1		2

to a consumer. It can be applied to nearly any value chain [53]. In fact, although VSM is often associated with manufacturing, the pertinent literature shows that it can be also successfully used in logistics, supply chain, service related industries, healthcare, software development, product development, administrative and office processes, and many more.

VSM allows identifying waste and its causes, providing a graphical mapping of all processes and activities of the logistics flow, helping to develop improving solutions [21].

The main elements which constitute the VSM procedure are usually the following [54]:

- (i) Identify the target product to map: firstly the value stream to be improved is determined, identifying the start and the end points.
- (ii) Draw the current state of the value stream map: the current state of material and information flows in the value stream are described, including any possible information (e.g., regarding current steps, tasks, costs, times, delays, etc.). To this aim, a team has to observe the manufacturing processes, documenting each fact (e.g., cycle times, buffer sizes, personnel requirements, etc.), and describing the system in an appropriate map by mean of standardized icons (see [55] for a complete list) in order to represent entities.
- (iii) Assess the current state map: the procedure investigates whether each process activity is adding value or not, thus allowing to identify the system anomalies. Each detected imperfections in the flow are signaled by means of a “bomb” icon.

3.3. Genba-Shikumi: quantify and rank the anomalies

After describing the warehouse by the UML and VSM tools, the proposed approach suggests, as schematized in Fig. 1, to analyze processes by means of the Genba Shikumi technique [32,51]. Genba is a Japanese term, literally meaning “the real place”, while Shikumi is the Japanese word which refers to something with different parts that are organized to work together under some intention. Hence, substantially Genba Shikumi stands for engineering method. In lean manufacturing, the main idea of Genba is that problems are visible, and the best improvement will come from directly going on the field to look for waste and opportunities. Hence, if a problem occurs, the management has to go where it occurs in order to thoroughly understand its impact, gathering data from all available sources. By applying the Genba philosophy, we consider the list of the main problems and anomalies, obtained in the previous VSM step and signaled by means of “bomb” icons.

On the basis of the Pareto principle, also known as the 80–20 rule, the law of the vital few, or the principle of factor sparsity, according to which for many events, roughly 80% of the effects come from 20% of the causes, we refine the results, assessing how each critical issue affects the overall value stream. In such a way we provide a mathematical formulation of the Genba philosophy. More in detail, we first compute a so-called *muda* matrix, shown in

Table 1, in which for each problem the type of waste and its cause are rated.

Denoting by p the number of bombs in the as-is state map and w the number of identified waste types and of their causes (man/method), the *muda* matrix $\mathbf{M} = \{m_{ij}\} \in \{0, 1\}^{p \times w}$ may be defined as a binary matrix whose generic element m_{ij} for $i = 1, \dots, p$ and $j = 1, \dots, w$ is as follows:

- $m_{ij} = 1$ if the i th identified problem has influence on the j th type of waste or error cause.
- $m_{ij} = 0$ if the i th identified problem does not have any influence on the j th type of waste or error cause.

Moreover, a muda vector $\mathbf{MV} = \{mv_i\} \in Z^p$ (see the example shown in Table 1) is calculated so that each mv_i is equal to the sum of the elements in the i th row of $\mathbf{M}$, i.e.:

$$mv_i = \sum_{j=1}^w m_{ij} \quad \text{for each } i = 1, \dots, p. \quad (1)$$

Hence, we define a correlation matrix $\mathbf{C} = \{c_{ij}\} \in \{0, 1\}^{p \times p}$, which is a binary square matrix of dimension p (equal to the number of detected bombs) where the generic element c_{ij} for $i, j = 1, \dots, p$ and $i \neq j$ is as follows:

- $c_{ij} = 1$ if the i th identified problem is correlated with the j th one.
- $c_{ij} = 0$ if the i th identified problem is not correlated with the j th one.

The generic diagonal element of $\mathbf{C}$ c_{ii} with $i = 1, \dots, p$ is set equal to zero.

Thus, $\mathbf{C}$ allows comparing in a pair wise manner each bomb assessing the degree of correlation between them. Hence, a correlation vector $\mathbf{CV} = \{cv_i\} \in Z^p$, an example of which is shown in the last column in Table 2, is determined such that each cv_i equals the summation of the elements in the i th row of $\mathbf{C}$, i.e.:

$$cv_i = \sum_{j=1}^p c_{ij} \quad \text{for each } i = 1, \dots, p. \quad (2)$$

Therefore, each $\mathbf{CV}$ element quantifies the correlation of each bomb with the others: the higher the correlation, the higher the element.

Subsequently, a so-called priority matrix $\mathbf{P} = \{p_{ij}\} \in Z^{p \times k}$ is computed, evaluating the impact of each identified issue on each of the k chosen performance indicators.

The generic element p_{ij} of $\mathbf{P}$ for $i = 1, \dots, p$ and $j = 1, \dots, k$ is defined as follows:

- $p_{ij} = 2$ if the i th identified problem has high impact on the j th chosen indicator.
- $p_{ij} = 1$ if the i th identified problem has low impact on the j th chosen indicator.
- $p_{ij} = 0$ if the i th identified problem has no impact on the j th chosen indicator.

Table 2
Example of correlation matrix and vector.

#	Observed problems	Problem 1	Problem 2	[...]	Problem p	CV
1	Problem 1		1			1
2	Problem 2	1		1		2
...	[...]		1		1	2
p	Problem p	1				1

Table 3
Example of priority matrix and vector.

#	Observed problems	KPI ₁	KPI ₂	[...]	KPI _k	PV
1	Problem 1	1	1	0	0	2
2	Problem 2	1	1	1	1	4
...	[...]	1	0	0	0	1
p	Problem p	0	0	2	1	3

Hence, a priority vector $\mathbf{PV} = \{pv_i\} \in Z^p$ is calculated, as in Table 3, so that each p_{vi} of $\mathbf{PV}$ is the summation of the elements in the i th row of $\mathbf{P}$, i.e.:

$$pv_i = \sum_{j=1}^k p_{ij} \quad \text{for each } i = 1, \dots, p. \quad (3)$$

The priority vector elements quantify how each waste impacts on the chosen performance indicators: the higher the priority, the higher the corresponding vector element.

Table 4 shows the so-called absolute importance vector ($\mathbf{AIV}$), obtained by summing up the three previously computed vectors:

$$\mathbf{AIV} = \mathbf{MV} + \mathbf{CV} + \mathbf{PV}, \quad (4)$$

Elements in the $\mathbf{AIV}$ vector quantify the severity of each evaluated criticism in the as-is state map: the greater such elements, the greater the need for solving the related problem.

Finally, we calculate a so-called area matrix, shown in Table 5, in which the $\mathbf{AIV}$ values are summed taking into account the area in which each issue occurs, thus allowing to identify the most problematic area, that are those requiring an immediate action.

Denoting by a the number of areas in the company, the area matrix is $\mathbf{A} = \{a_{ij}\} \in Z^{p \times a}$, and the area vector is $\mathbf{AV} = \{av_i\} \in Z^a$, so that each av_i is the sum of the elements in the i th column of $\mathbf{A}$, i.e.:

$$av_i = \sum_{j=1}^p a_{ij} \quad \text{for each } i = 1, \dots, a. \quad (5)$$

Table 4
Example of absolute importance vector.

#	Observed problems	MV	CV	PV	AIV
1	Problem 1	1	1	2	4
2	Problem 2	2	2	4	8
...	[...]	1	2	1	4
p	Problem p	2	1	3	6

Table 5
Example of area matrix and vector.

#	Observed problems	Area 1	Area 2	[...]	Area a
1	Problem 1	6			2
2	Problem 2	3		2	
...	[...]	6	6		1
p	Problem p	2		4	5
	AV	17	6	6	8

3.4. VSM-UML-Reengineering: verify, update and reengineer the redesigned warehouse

At this point the future state map is drawn. Hence, in this step the warehouse is re-designed eliminating (at least a certain number of) the non-value adding activities in the descending order of importance obtained from the previous step. Hence, the ideal future state of the system is drawn, showing the desired way in which processes should look in the future after eliminating the (top) detected criticalities.

Subsequently, the UML framework is reapplied to detail the warehouse activities according to the redesigned state map.

Finally, the identified improvement plans are implemented in the warehouse, so as to obtain the desire future state.

Obviously, to ensure a continuous improvement and to verify the implementation plans, the proposed approach for warehouse analysis and optimization is continuously iterated, e.g., at least once a year.

3.5. Discussion of the proposed approach with respect to the literature

We conclude this section by a discussion of the merits of the proposed lean warehousing technique with regard to the related literature.

With respect to the classical VSM technique for manufacturing systems optimization, the proposed procedure offers several advantages: (i) it can be easily applied even to complex warehousing systems (while VSM is typically applicable to linear

flows systems, see e.g. [1]); (ii) it leads at the same time to detailing the process activities with their players and responsibilities (thanks to the UML formalism) and to depicting the overall warehouse in a high level picture (using the VSM synthesis ability); (iii) it allows the quantitative assessment of the detected system anomalies (thanks to the Genba Shikumi methodology).

Moreover, with respect to the existing literature contributions related on the warehouse management topic, e.g., the recalled papers by Gagliardi et al. [18] and Kofjac et al. [49], it is to be remarked that the presented approach is systematic, general, and is based on the lean philosophy of reducing waste.

In addition, comparing the presented methodology with the few contributions in the related literature we remark the following issues. With respect to the approach proposed by Gopakumar et al. [29], our methodology benefits from the use of the UML framework in order to detail the optimized warehouse configuration and from the mathematical formalization of the Genba Shikumi philosophy for a synthetical and yet effective evaluation of the warehouse criticalities, while avoiding complex simulation that require a very deep data investigation and analysis campaign in order to be effective. Moreover, with respect to the approach by Chen et al. [30], our methodology, thanks to the integration of UML and Gemba Shikumi with VSM, leads to a more systematic and general technique to detail the optimized warehouse. Furthermore, comparing the presented approach with the recalled contribution by Dharmapriya and Kulatunga [28] we remark that, while our methodology is focused on optimizing the warehouse by identifying, ranking, and eliminating waste, their approach is rather addressed at the layout design problem in a lean context.

4. The case study

In this section we present an application of the approach to the case study. The company is an Italian producer of interior design objects, originally founded as a craft business cabinet that has recently strongly increased its worldwide market. Its logistics is

currently organized with a low-tech and low information contents, and currently generates many wastes, hence the firm plans to set up an integrated and automated logistics platform to match its needs of growth and to reduce anomalies. The internal logistics are currently managed mainly manually by the staff supported by a basic information system. The warehouse management is mainly based on a shared storage strategy, hence, when one location becomes empty, it may be reassigned even to a different kind of product. This leads to a high utilization of space, but at the same time, since products change their location over time, it is time-consuming for workers to search for them. For these reasons, it is apparent that an effective warehouse management strategy would require a software support and more disciplined warehouse processes.

The warehouse flows are as follows. Goods arrive at the warehouse following two main streams: (1) from the outside, with the supply of wood panels, silver, glue, and various decorations for the assembly of finished products, and (2) from within the company, after the working of wood panels and/or their customization and/or final assembly. The wood panels stock is arranged outside the company, while a semi-finished products warehouse is located inside the company. Both are managed without any ICT support. The current company database includes more than 20 000 products, all fully customizable, which is why a warehouse of finished products is not present.

According to the first step of the lean warehousing approach in Fig. 1, we use UML in order to represent in detail the system activities and highlight the respective actors and responsibilities.

Fig. 2 shows that in the company orders are handled based on their dimension: hence, two different flows may be singled out. Particularly, in case of large orders (that is, more than 100 pieces), the “Sales and Logistics Office” registers them by assigning to each one a priority, that derives from the planned delivery date, as agreed with the customer. The record ends with the delivery of the picking list and the order requirements to the “Design Office”. Here, with the help of a CAD software, the cutting of the wood

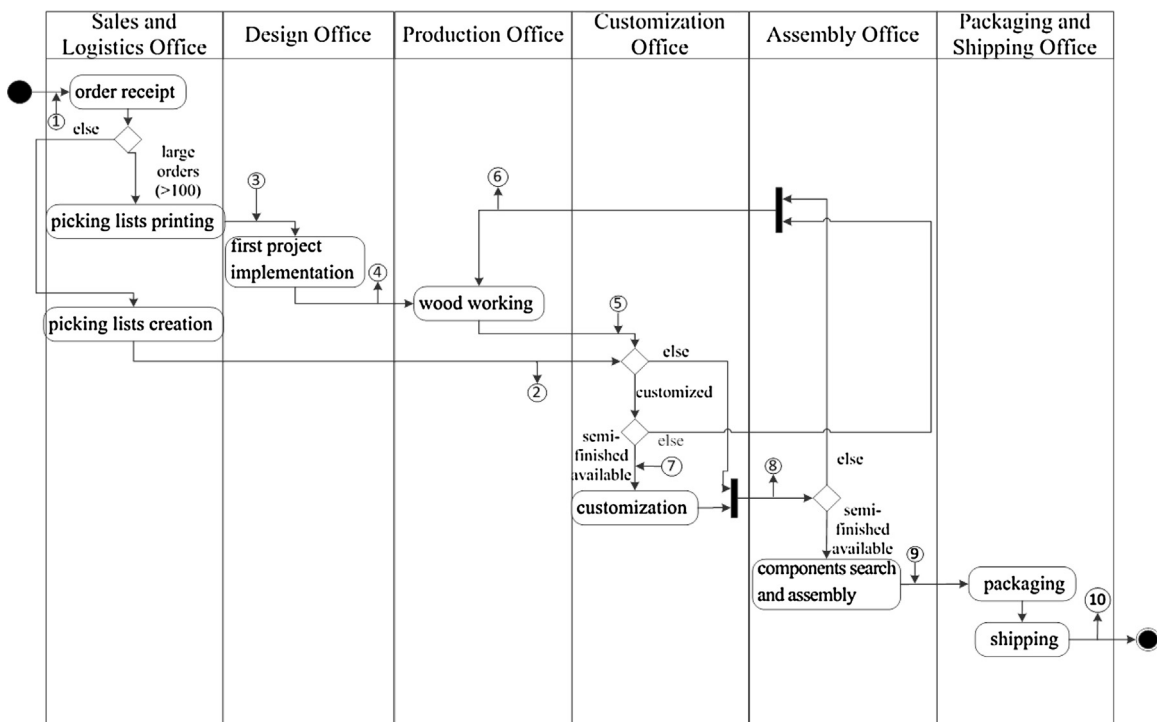


Fig. 2. Activity diagram of the company logistics (as-is).

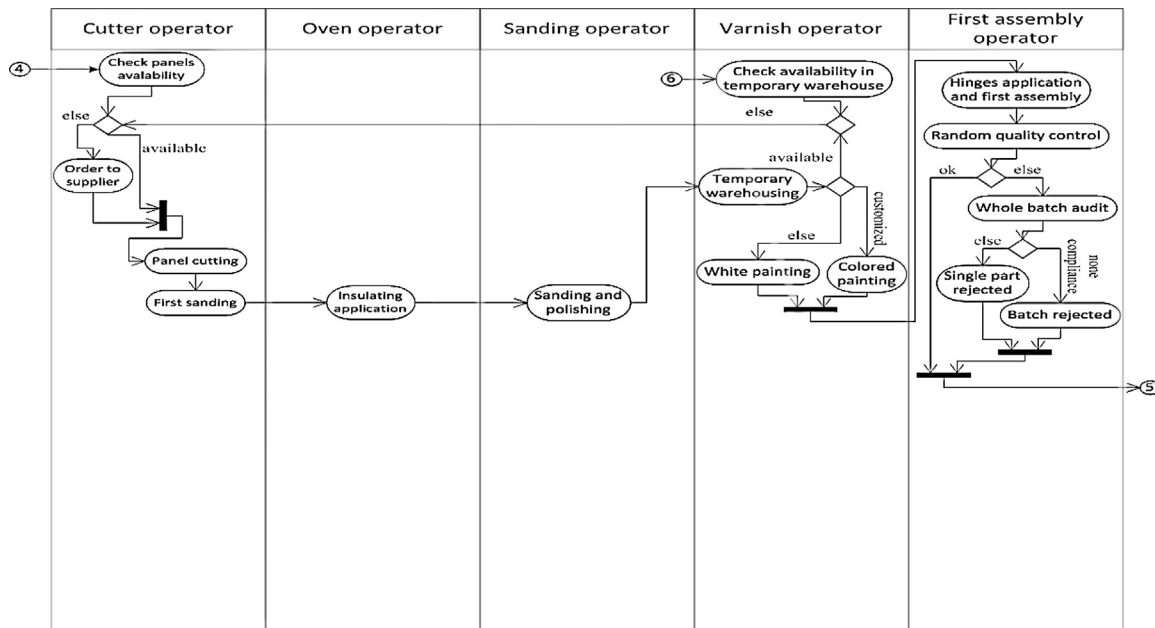


Fig. 3. UML diagram of the wood working activity 4–5 of Fig. 2 (as-is).

panels to fulfill the order is planned. Subsequently, panels are cut with a numerically controlled milling machine in the “Production Office”, then they are customized with prints or laser engravings in the “Customization Office”, and the production process ends with the delivery of the finished customized pieces to the “Assembly Office”, where it is applied the requested decoration (silver, glass, etc.). Finally, assembled products are delivered to the “Packaging and Shipping Office”, where they are packaged and an identification code is applied for the final delivery to an external carrier.

On the other hand, in case of small orders, the “Sales and Logistics Office” collects them together with similar orders, creating the order list which is delivered to the “Customization Office”. Here the availability of the required semi-finished products is checked. If these are available, the required customization is performed, followed by the assembly process, with the application of the requested decoration, and finally the products are packaged with the identification code application and delivered to the external carrier, as seen before. Otherwise, in case semi-finished products that are necessary to complete the order are not immediately available, the number of needed pieces is forwarded to the “Production Office”. Here the availability of the required components is checked in the temporary warehouse, or otherwise their production is started. Then the needed pieces are delivered to the “Customization Office”, where they follow the same previous flow.

Among all the activities described above, the one that mostly interfaces with the warehouse is wood working (that is the activity between labels 4 and 5 in Fig. 2), hence its current UML activity diagram is shown in Fig. 3.

In the production phase depicted in Fig. 3, the cutter operator checks for the availability of the wood panels that are needed to complete the order. In case they are unavailable or present in limited quantities, he sends an order to the “Logistics Office”. Once obtained the panels, he loads the cut project on a numerically controlled milling machine and launches the cutting program. When the cut is over, panels are grossly sanded, in order to separate pieces and obtain a predetermined desired thickness. Afterwards panels are put into an oven, where a layer of filler and insulating material is applied. Then the pieces are polished and sanded to be placed in the temporary warehouse,

becoming available in case of need. Subsequently, jointly with other items related to unexecuted orders, it is applied a background of white paint or color, depending on the type of desired customization. Hence, hinges, nails and anything else necessary to a first assembly are applied, and finally a random quality control is made. Whether non-compliance is detected, the entire batch is fully checked, so that all non-compliant pieces are produced again. On the other hand, pieces that pass the inspection are loaded onto the internal lift to be transported in the “Customization Office”.

Having detailed the company flows, by the second step in Fig. 1 we identify in detail all the problems and anomalies occurring in the internal logistics, with the aim of identifying the sources of errors and inefficiencies and removing them. The main purpose is to describe all the existing interactions between the parts composing the warehouse processes, in order to standardize and optimize them. To this aim, the proposed lean warehousing approach uses the VSM technique, a fundamental tool of the so-called lean philosophy, which, in the case study, allows obtaining the overall current state map in Fig. 4, where the actors and supply policies are graphically depicted and the legend describes symbols. The figure shows that components that flow through the warehouse are composed by packages coming from the internal or external production that are stored in the warehouse, prepared due to an order, and finally delivered to the carrier.

Basically, the warehouse represents the decoupling point between the main activities of production and distribution. In fact, it receives incoming goods from the “Production Office” according to a push policy (labeled by the so-called “Push Arrow Icon” in Fig. 4), while they are stored and retrieved in presence of an order following a pull strategy (labeled by a “Pull Arrow Icon” in Fig. 4). More precisely, if an order is present, then all the required goods are moved to the loading bay, where an inventory structure (signaled by a “Supermarket Icon” in Fig. 4) is available to pickers. Here goods verification and truck loading are performed.

After describing the company internal logistics and warehouse by means of UML and VSM, processes are analyzed by means of the proposed mathematical formulation based on the Genba Shikumi philosophy and on the Pareto principle.

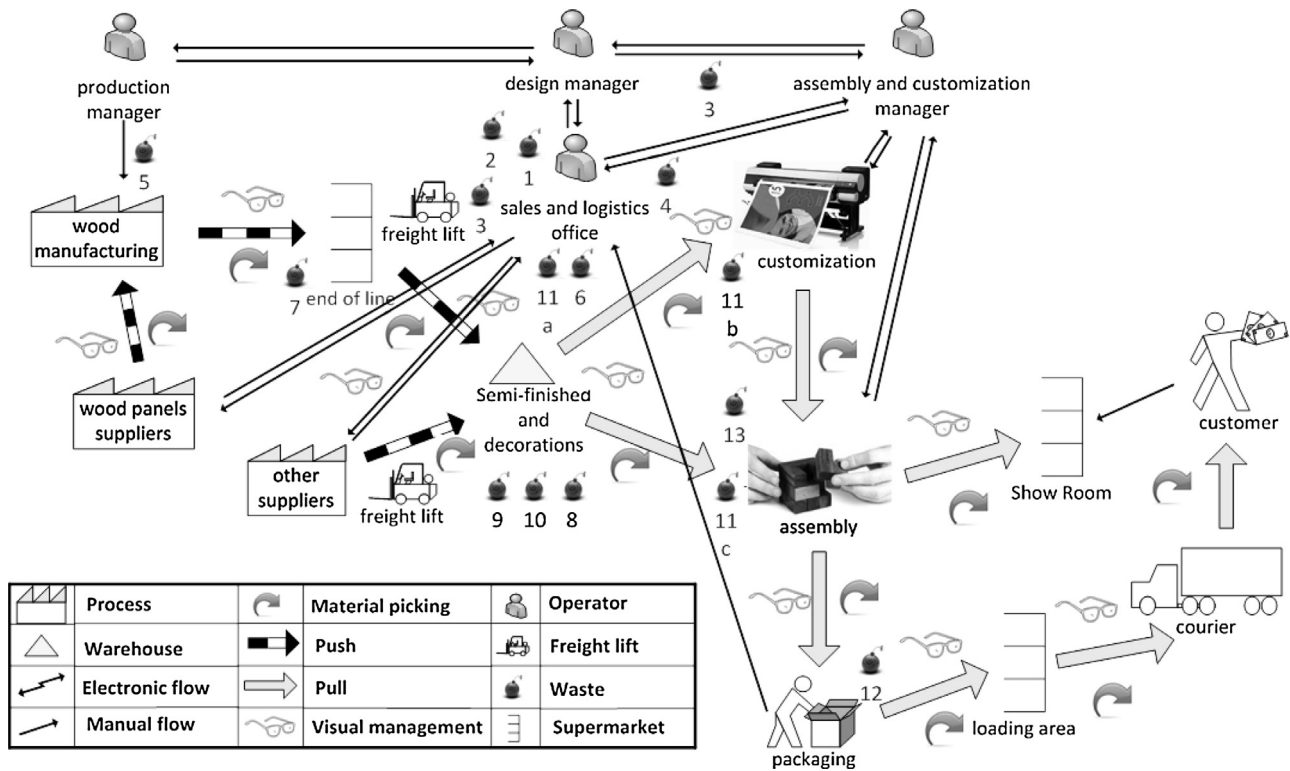


Fig. 4. Overall VSM current state map (as-is).

We first compute the *muda* matrix and the *muda* vector, shown in Table 6, in which for each problem the type of waste (i.e., overproduction, underproduction, movements, space, waiting time, safety, over processing flows) and its cause (i.e., manual error or type of method adopted) are rated.

The *muda* matrix clearly shows that the current problems in the case study warehouse are generated by a lack of ICT and automation, leading above all to wasted time, under production and unnecessary movements, as shown by the values in the last row in Table 1. Furthermore, it is clear that the lack of methods and standardized procedures are relevant for the examined company. Hence, we define the correlation matrix and the correlation vector to compare each bomb assessing the degree of correlation between them, as shown in Table 7.

The CV values in Table 7 show that the problems of lack of communication with the “Assembly Office” and with the “Customization Office” are linked with eight other issues and that the un-replaceable staff is linked to six among the other identified problems.

Subsequently, the priority matrix and the priority vector are computed and reported on Table 8, evaluating the impact of each identified issue on each of the $k = 3$ chosen performance indicators. These include both tangible or numerically measurable KPIs – namely the errors percentage, i.e., the percentage of wrong lines in an order, and profitability, measured in terms of the number of shipped parcels per hour – and an intangible KPI, i.e., the quality perceived by the customer, detected by means of customer satisfaction interviews.

Table 6
The muda matrix and vector.

#	Observed problems	Overproduction	Underproduction	Movements	Space	Waiting time	Safety	Overprocessing	Man	Method	MV
1	Lists creating					1				1	2
2	Order entry			1	1	1			1	1	5
3	Database updating					1			1		2
4	Checking the availability of external suppliers		1			1				1	3
5	Communication with the production office	1	1	1		1				1	5
6	Checking the availability of woods panels		1			1				1	3
7	Checking the availability on the temporary warehouse	1		1		1			1	1	5
8	Checking the availability of white semifinished	1	1	1		1		1	1	1	7
9	Checking the availability of other semifinished	1	1	1		1		1	1	1	7
10	Difficulty in finding semifinished and decorations	1	1	1		1		1	1	1	7
11	Non-replaceable staff (11a–11c)		1			1		1		1	4
12	Correspondence check of codes					1			1	1	2
13	Communication with the assembly and customization offices		1			1			1	1	4
		5	8	6	1	12	0	4	8	12	

Table 7

The correlation matrix and vector.

#	Observed problems	Lists creating	Order entry	Database updating	Checking the availability of external suppliers	Communication with the production office	Checking the availability of woods panels	Checking the availability on the temporary warehouse	Checking the availability of white semifinished	Checking the availability of other semifinished	Difficulty in finding semifinished and decorations	Non replaceable staff (11a–11c)	Correspondence check of codes	Communication with the assembly and customization offices	CV
1	Lists creating		1									1			2
2	Order entry	1		1								1			3
3	Database updating		1										1	1	3
4	Checking the availability of external suppliers										1			1	2
5	Communication with the production office						1	1						1	3
6	Checking the availability of woods panels					1		1	1	1					4
7	Checking the availability on the temporary warehouse					1	1					1		1	4
8	Checking the availability of white semifinished						1					1		1	3
9	Checking the availability of other semifinished						1					1		1	3
10	Difficulty in finding semifinished and decorations				1							1		1	3
11	Non-replaceable staff (11a–11c)	1	1					1	1	1	1				6
12	Correspondence check of codes			1										1	2
13	Communication with the assembly and customization offices			1	1	1		1	1	1	1		1		8

Table 8

The priority matrix and vector.

#	Observed problems	Error %	Profitability	Quality	PV
1	Lists creating	1	1	0	2
2	Order entry	1	1	1	3
3	Database updating	1	0	0	1
4	Checking the availability of external suppliers	0	0	1	1
5	Communication with the production office	2	2	0	4
6	Checking the availability of woods panels	0	0	1	1
7	Checking the availability on the temporary warehouse	0	0	1	1
8	Checking the availability of white semifinished	0	1	0	1
9	Checking the availability of other semifinished	0	1	0	1
10	Difficulty in finding semifinished and decorations	2	2	0	4
11	Non-replaceable staff (11a–11c)	1	2	0	3
12	Correspondence check of codes	2	2	0	4
13	Communication with the assembly and customization offices	0	1	0	1

Table 9

The absolute importance vector.

#	Observed problems	MV	CV	PV	AIV
1	Lists creating	2	2	2	6
2	Order entry	5	3	3	11
3	Database updating	2	3	1	6
4	Checking the availability of external suppliers	3	2	1	6
5	Communication with the production office	5	3	4	12
6	Checking the availability of woods panels	3	4	1	8
7	Checking the availability on the temporary warehouse	5	4	1	10
8	Checking the availability of white semifinished	7	3	1	11
9	Checking the availability of other semifinished	7	3	1	11
10	Difficulty in finding semifinished and decorations	7	3	4	14
11	Non-replaceable staff (11a–11c)	4	6	3	13
12	Correspondence check of codes	2	2	4	8
13	Communication with the assembly and customization offices	4	8	1	13

By analysing Table 8 it can be observed that the problem of verifying the correspondence between the codes has a strong weight in the company production processes, as well as finding semi-finished and decorations in the warehouse, and the lack of communication with the Production Office.

Table 9 shows the absolute importance vector (AIV), which quantify the severity of each evaluated criticism in the as-is state map.

Hence, we calculate the area matrix, shown in Table 10, in which the AIV values are summed taking into account the area in which each problem occurs, thus allowing to identify the most problematic area, that are those requiring an immediate action.

As it is clear from Table 10, the “Assembly Office” and the “Customization Office” are the areas characterized by the greatest

waste, while, among these, those numbered 10, 11, 13 (in a descending order of importance) require an urgent action plan. In particular, the search of semi-finished and decorations in the warehouse results to be the most severe problem, hence, to solve this issue and all those related to it (as evaluated by the correlation matrix), in the subsequent section we propose to set up a reengineered warehouse.

The subsequent step of the lean warehousing approach, as depicted in Fig. 1, is to remove the most relevant identified anomalies. To this aim, since the main problems are finding semi-finished parts and decorations in the warehouse and the lack of communication between the various business offices, and the areas characterized by the greatest waste are the “Assembly Office” and the “Customization Office”, we re-design the company

Table 10

The area matrix and vector.

#	Observed problems	Sales and logistics office	Design office	Production office	First assembly and quality control office	Customization office	Assembly office	Packaging and shipping office
1	Lists creating	6						
2	Order entry	11						
3	Database updating	6	6			6		
4	Checking the availability of external suppliers	6					6	
5	Communication with the production office	12			12	12	12	
6	Checking the availability of woods panels			8				
7	Checking the availability on the temporary warehouse			10				
8	Checking the availability of white semifinished					11		
9	Checking the availability of other semifinished						11	
10	Difficulty in finding semifinished and decorations					14	14	
11	Non-replaceable staff (11a–11c)	13	13	13		13	13	
12	Correspondence check of codes							8
13	Communication with the Assembly and Customization Offices	13			13	13	13	
AV		67	19	31	25	69	69	8

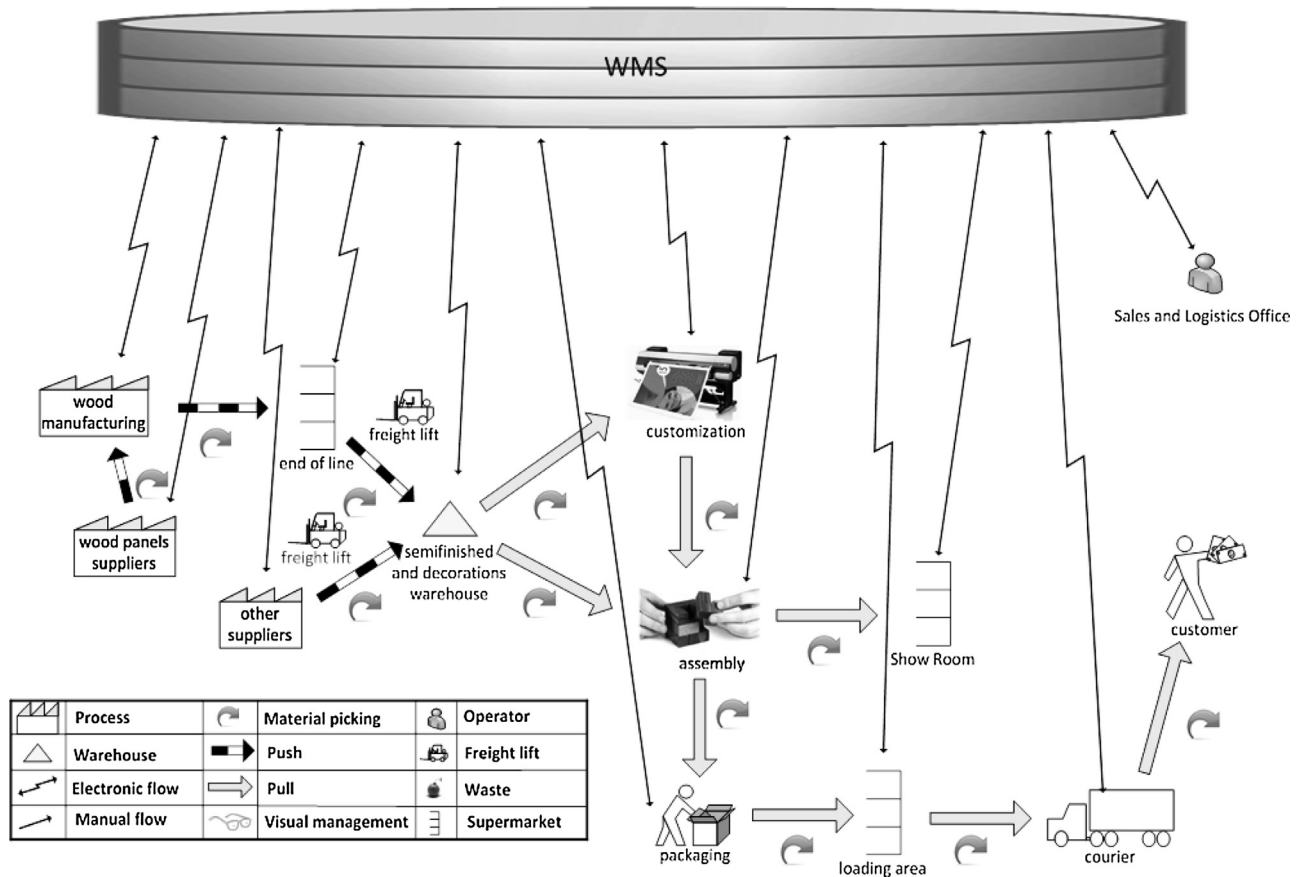


Fig. 5. Overall VSM future state map (to-be).

warehouse. Hence, we propose to adopt sliding compact shelves, and, based on the strategic importance of the ICT solutions in lean manufacturing, to implement an automated control of the picking area and pickers paths by a dedicated warehouse management system (WMS).

A WMS is mainly an information and communication technology (ICT) system that is used to physically control the warehouse database [56], thus supporting the daily operations within the

warehouse, helping to manage inventory, storage locations and workforce. It primarily aims at supporting and monitoring the movement and storage of materials within a warehouse and processing the associated transactions, including inbound processes (receiving and put-away), and outbound processes (order picking, packing and shipping), as well as returns. A WMS also directs and optimizes stock put-away based on real-time information about the status of utilization.

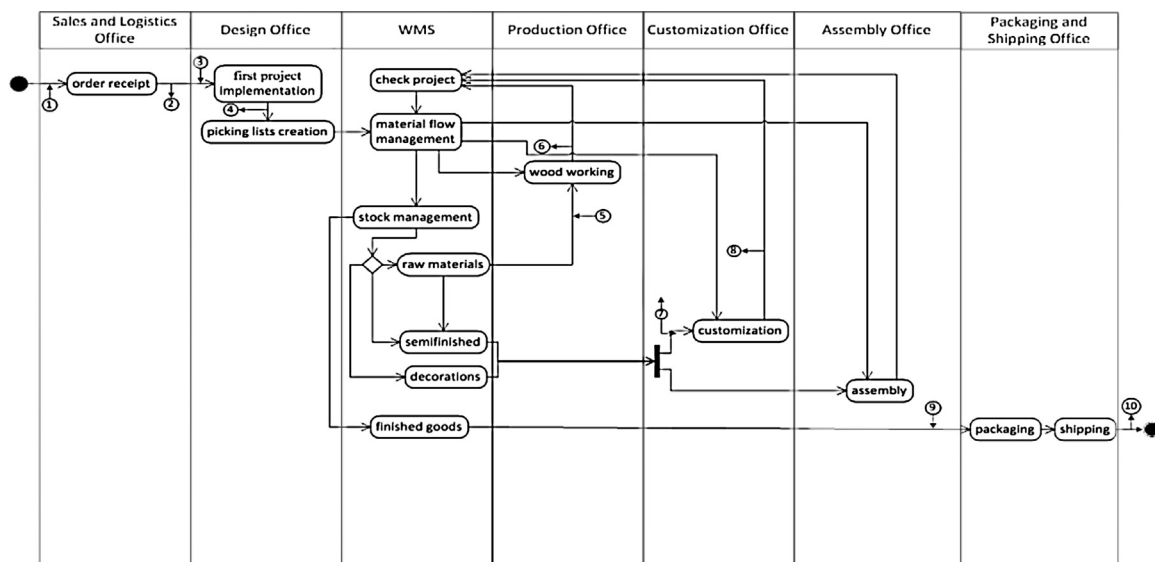


Fig. 6. Activity diagram of the company logistics (to-be).

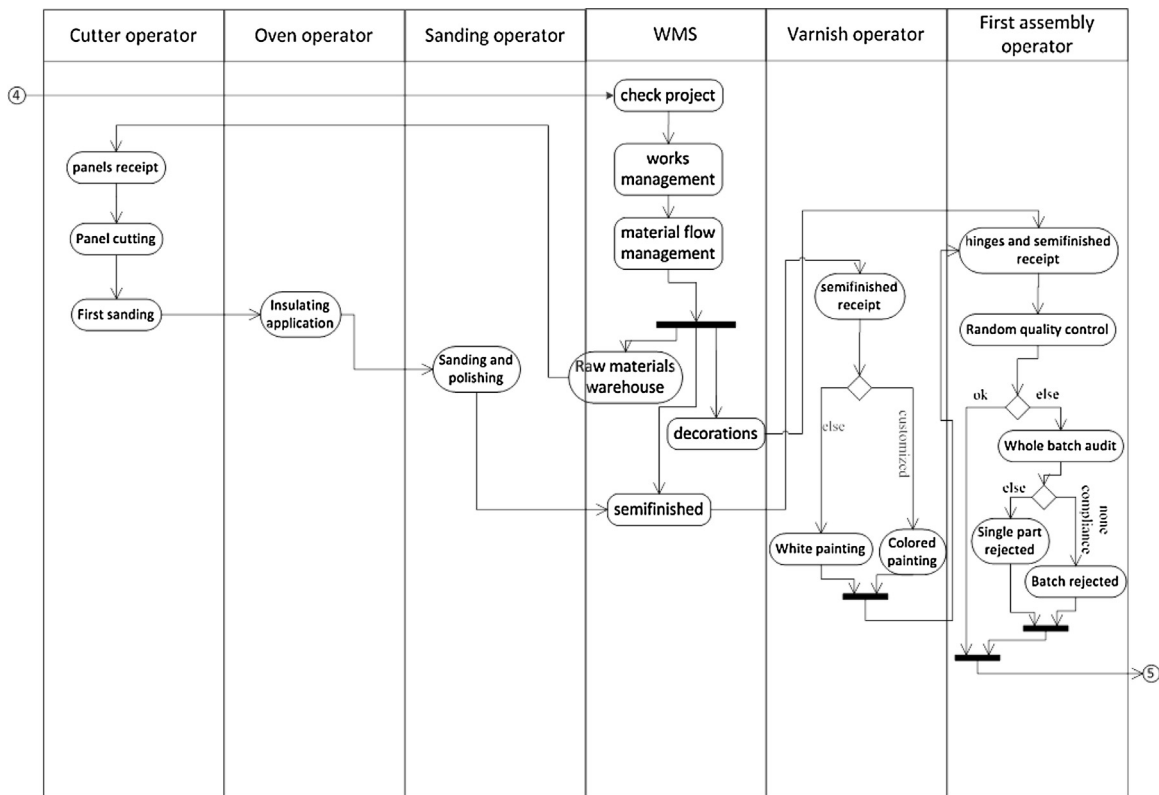


Fig. 7. Activity diagram of the wood working 4–5 activity of Figure 6 (to-be).

Hence, we obtain the future state map of the proposed solution, realizing that it leads to eliminating all bombs and manual controls, thus reducing the error probability and increasing profitability. Fig. 5 depicts the to-be state map, showing how the WMS coordinates both material and information flows, thus allowing to reduce the errors due to human intervention or derived from the adopted method.

Subsequently, it is necessary to develop the UML model of the future state map including the proposed WMS solution. Fig. 6 shows the to-be UML macro activity diagram and Fig. 7 depicts the to-be UML production activity.

Comparing the UML to-be diagrams in Figs. 6 and 7 with the corresponding as-is diagrams depicted in Figs. 2 and 3, it can be remarked once again that most of the flows, both of information and material type, are coordinated by the WMS in the reengineered warehouse, thus demonstrating its contribution in their management and the resulting possibility to reduce errors.

Since there are neither available data on competitor warehouses nor a historical database on customer orders, to judge the effectiveness of the redesigned warehouse, we compare it with the actual warehouse, hence fixing the warehouse geometry and layout, and the freight mix, in order to obtain absolute data that are directly evaluable. The data of the as-is state and the estimated ones in the to-be state (i.e., by adopting sliding compact shelves and WMS) are illustrated in Table 11, while Table 12 shows the obtained value of some suitable performance indices. In particular, Table 12 shows how the reengineered solution implementation has led to about a 25% average increase in the occupied stock volumes and a 30% average increase in the available free surface (i.e. the surface that is not occupied by shelves and material on the floor, and is not required for handling), hence decreasing the warehouse saturation, with a corresponding increase in organizational efficiency and chances of larger stocks. Note that the improvements underlined in Table 12 are connected to the

adoption of sliding shelves, since those due to the WMS implementation are related to the information exchange and are not yet quantified. Hence, the improvement is a well conservative estimate.

Obviously, the implementation requires a KPI monitoring of the novel solution, so as to estimate how much this action plan corresponds to expectations (reduction of error probability, increase of profit and service quality perceived by customers). Hence, the firm has to start a warehouse activity profiling (including an ABC analysis in order to know what stock units mostly matter, or determine the Pareto curves in order to rank the stock units by popularity, e.g. the number of requests), with a careful measurement and statistical analysis of warehouse activity and on customer orders, and how these determine the workload within the facility. It is also important to understand how the intensity of work varies over time, since most products have some natural cycle or seasonality that repeats over the year (e.g., for Christmas products). Note that all data required to support profiling, that is those pertaining to each stock unit, those pertaining to customer orders, and those related to stock locations within the warehouse, may be easily collected by the WMS. Clearly, applying in a recursive way the proposed approach, the obtained improvements can be further identified and quantified.

Table 11
Warehouse data.

Data	Meaning	as-is	to-be
V_s	Available shelves volume [m ³]	300	376
S_s	Surface occupied by shelves [m ²]	40	50
S_h	Handling area [m ²]	120	130
S_f	Area occupied by materials on the floor [m ²]	52	–
S_w	Warehouse area [m ²]	320	320

Table 12

Warehouse performance indices.

Index	Meaning	as-is	to-be	Improvement
$I_1\% = S_s/S_w$	Surface occupied from the shelves over the total	12.50%	15.63%	25.00%
$I_2\% = [V_s(\text{to_be}) - V_s(\text{as_is})]/V_s(\text{as_is})$	Increase in the available volume	–	25.33%	25.33%
$I_3\% = (S_w - S_f - S_s)/S_w$	Total available free surface	71.25%	84.38%	18.42%
$I_4\% = (S_w - S_f - S_s - S_h)/S_w$	Free surface	33.75%	43.75%	29.63%

5. Conclusions

This paper presents a lean warehousing approach for the strategic design and reengineering of production warehouses. The proposed approach integrates the Unified Modeling Language (UML), Value Stream Mapping, and Genba Shikumi methodologies. Their combination allows detecting the critical issues of a warehouse, assessing them quantitatively, and identifying the most appropriate response actions to achieve the desired improvement. In particular, the UML formalism is adopted in order to describe and analyze the internal logistic, warehouse activities and the corresponding responsibilities, and the VSM tool is used in order to highlight in which area any possible waste is located. The application of the Genba Shikumi philosophy is adopted in order to further characterize waste, and thanks to a novel mathematical formulation based on the Pareto principle, to rank their importance. The iterative nature of the presented integrated approach brings to the continuous improvement which is central to the lean philosophy.

The application of the proposed technique to an Italian producer of interior design objects shows its simplicity and effectiveness, highlighting that the problems affecting the company warehouse and inferring that they are caused by the absence of an integrated and automated warehouse management. Using the proposed approach, we show that the adoption of a warehouse management system leads to improvements in the production warehouse performance. Future investigations will deal with enhancing the company warehouse reengineering by a widely used solution for warehouse material handling, characterized by high accuracy, high speed and high storage capacity, i.e., automated storage and retrieval systems or carousels, hence embedding the warehouse control system. Moreover, once historical data are available, a comparison with competitors will be possible, e.g., by using a cross-efficiency data envelopment analysis approach.

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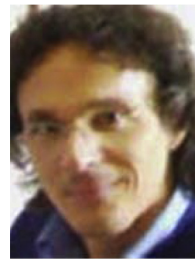


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